The midterm exam covers Module 1, 2 and 3, as well P1 in Go, and P2 in Scheme.

Module and programming language, there is vocabulary and associated concepts for you to understand. I recommend starting with this to see if you can describe as many of the elements from the list below as you can.

Module 1:

Vocabulary:

- Computer
- Von Neumann architecture
- Imperative Language
- Declarative Language
- Translator
- Compiler
- Interpreter
- Hybrid Translator
- Frontend
- Backend
- Scanner
- Parser
- Symbol Table

Concepts:

- Programs as Data
- Why are there so many languages?
- What makes a programming language successful?
- Phases of Compilation
- Phases of Interpretation
- Frontend components and purpose
- Bootstrapping purpose

Module 2:

Vocabulary:

- Alphabet
- String/Lexeme
- Language
- Syntax
- Semantics
- Regular Expressions / RE
- Context Free Grammar / CFG
- Ad-hoc Scanner
- Table Driven Scanner
- Backus-Naur Form
- Ambiguity
- Terminal Symbol
- Non-terminal Symbol
- Production
- Expansion
- Derivation
- Left-most Derivation
- Right-most Derivation
- Parse Tree
- Parser
- LL/predictive parser
- LR/shift-reduce parser

Concepts:

- What are the 5 basic rules of Regular Expressions
- What are limitations of Regular Expressions
- How does a Scanner/Lexer work? What are its critical components?
- How does a Scanner/Lexer respond to errors?
- What makes a CFG ambiguous? How can you prove it?

- Be comfortable creating parse trees for any given CFG.
- Why is left recursion in a CFG problematic for a LL parser?
- How are precedence and associativity related to parse trees?
- How does a recursive descent parser map to a CFG?
- How does a parser respond to errors?

Module 3:

Vocabulary:

- Name
- Binding
- Lifetime
- Scope
- Stack
- Heap
- Fragmentation
- Static Scope
- Dynamic Scope
- Closure
- Shallow Binding
- Deep Binding
- Shadow
- Override
- Alias
- Override
- Overload
- Polymorphism
- Subtype Polymorphism
- Parametric Polymorphism
- Namespace

Concepts:

- Computer memory organization

Go Programming Language

Vocabulary

- Receiver
- Channel
- goroutine

Concepts

- Structure of a program
- Compositional inheritance
- Channels
- Functions vs. Methods

Scheme Programming Language

Vocabulary:

- car
- cdr
- cons

Concepts

- List construction and deconstruction
- Prefix form
- Recursive Algorithms
- Be able to read and interpret short programs in scheme